

GAUS AG on
D-elliptic sheaves and Hasse principle

NOTES FOR TALK 3 on
STACKS II

§ 1. BRIEF RECAP OF TALK 1
(for more details, refer to the
notes for Talk 1 by Prof. Yu)

Generally moduli functors

$F: \text{Sch}_k^{\text{op}} \longrightarrow \text{Sets}$ are not
representable by schemes, due to
existence of non-trivial automorphisms
of objects being classified by F .

Ex. $M_{1,1}: \text{Sch}_{\mathbb{C}}^{\text{op}} \longrightarrow \text{Sets}$
 $T \longmapsto \{ \text{Elliptic curves over } T \} / \sim$

is not representable by a scheme,
for instance, due to non-injectivity of

$$M_{1,1}(\text{Spec}(\mathbb{C}(t))) \longrightarrow M_{1,1}(\text{Spec}(\mathbb{C}(t^{1/6})))$$

where two elliptic curves $Y^2Z = X^3 - tZ^3$
and $Y^2Z = X^3 - Z^3$ are non-isom. over
 $\mathbb{C}(t)$ but isomorphic over $\mathbb{C}(t^{1/6})$.

To remedy this situation of non-representability, one allows the isomorphisms of objects to be recorded, thereby forming a 2-functor

$$M_{1,1}: \text{Sch}_{\mathbb{C}}^{\text{op}} \rightarrow \text{Grpd}$$

$$T \mapsto \langle \text{Elliptic curves over } T \rangle$$

This forms a **stack**. To avoid 2-categorical language, one instead forms a **category fibered on groupoids (CFG)**

$$P: \mathcal{M}_{1,1} \rightarrow \text{Sch}_{\mathbb{C}}$$

$$(T, \text{Elliptic curve } E/T) \mapsto T$$

The defining condition for a stack $P: \mathcal{M} \rightarrow (\text{Sch}_k)_{\text{fppf}}$ is that

- (i) $\forall T \in \text{Sch}_k$ & $x, y \in \mathcal{M}(T)$, the presheaf $\underline{\text{Hom}}(x, y)$ on T is a sheaf;
- (ii) Descent data is effective.

These two conditions can be

summed up as an equivalence of categories

$$\mathcal{M}(T) \longrightarrow \mathcal{M}(T' \xrightarrow{f} T)$$

$$x \longmapsto f^*x$$

where $T' \xrightarrow{f} T$ is a fppt, surjective morphism and $\mathcal{M}(T' \xrightarrow{f} T)$ is a category naturally encapsulating descent data w.r.t. $T' \xrightarrow{f} T$.

(YONEDA) An equivalence $\text{Hom}_{\text{Stacks}}(\underline{T}, \mathcal{M}) \simeq \mathcal{M}(T)$

GEOMETRY ON STACKS

DEF'N A stack $p: \mathcal{M} \rightarrow \text{Sch}_k$ is said to be Deligne-Mumford (DM) if

(i) the diagonal $\mathcal{M} \rightarrow \mathcal{M} \times \mathcal{M}$ is representable by schemes, and is quasi-compact and separated.

$\Leftrightarrow$ (i') $\underline{\text{Hom}}(x, y)$ is rep'ble by a quasi-compact and separated scheme $\forall x, y \in \mathcal{M}(T)$

$\Rightarrow$ (i'') Any $\underline{X} \rightarrow \mathcal{M}$ is rep'ble by scheme and is quasi compact and separated

(ii) $\exists$ a k -scheme U and an étale, surjective $U \rightarrow \mathcal{M}$ (U is called an atlas)

Ex. $\mathcal{M}_{1,1}$ is a DM-stack.

Proving it's a DM stack is quite non-trivial. We'll see below another example of a DM stack, namely quotient stacks, where it's easier to show this.

§2. QUOTIENT STACKS

DEF'N: ((PRINCIPAL) G -BUNDLE) Let Y be an S -scheme, G a smooth, affine Y -group scheme. Then a (principal) G -bundle is a pair of morphisms $(\pi: P \rightarrow Y, \rho: G \times_Y P \rightarrow P)$ s.t. π is flat, locally finitely presented and surjective, and

$$(i) \begin{array}{ccc} G \times_Y G \times_Y P & \xrightarrow{(id, \rho)} & G \times_Y P \\ (m, id) \downarrow & \square & \downarrow \rho \\ G \times_Y P & \xrightarrow{\rho} & P \end{array} \text{ commutes ;}$$

(ii) $P \xrightarrow{(e\pi, id)} G \times_Y P \xrightarrow{p} P$ is identity

(iii) $G \times_Y P \xrightarrow{(p, pr_2)} P \times_Y P$ is an isom.

Example $\div$ (Trivial G -bundle)

$(G \rightarrow Y, p = m: G \times_Y G \rightarrow G)$

Fact $\div$ Every G -bundle is étale locally trivial.

DEF'N $\div$ (QUOTIENT STACK) Let

X/k be a scheme, G/k a smooth, affine group scheme. Then the quotient stack $[X/G] \rightarrow (\text{Sch}_k)_{\text{fppf}}$ is the CFG s.t. for any $T \in \text{Sch}_k$,

$$[X/G](T) \doteq \left(\begin{array}{l} P \rightarrow T \text{ a } G_T\text{-bundle,} \\ P \rightarrow X \text{ a } G\text{-equivariant} \\ \text{map} \end{array} \right)$$

For G étale, $[X/G]$ is a DM stack

-Effectiveness of descent data

This follows from the equivalence
 $\{G\text{-bundles}\} \longleftrightarrow \{G\text{-torsors}\}$
 and the effectiveness of
 descent data for the
 stack of sheaves.

↑ certain
 sheaves
 with a
 suitable
 action of h_G

- Representability of the diagonal

Sufficient to show that for scheme
 T and $P_1, P_2 \in [X/G](T)$, the
 sheaf Hom (P_1, P_2) is represen-
 table by a scheme. Enough to
 show this étale locally, where there
 are isomorphisms $\sigma_1: P_1 \cong G_T, \sigma_2: P_2 \cong G_T$,
 if we have $p_1: G_T \rightarrow X_T, p_2: G_T \rightarrow X_T$.
 the corresponding G -equivariant maps
 Consequently, Hom (P_1, P_2) is rep'ble by
 the fiber product ?,

$$\begin{array}{ccc} ? & \xrightarrow{\quad} & G_T \\ \downarrow & \lrcorner & \downarrow (p_1(e), p_2) \\ X_T & \xrightarrow{\Delta} & X_T \times X_T \end{array}$$

- An étale, surjective atlas

Let $\underline{X} \rightarrow [X/G]$ be the map corresponding to the trivial G -bundle. Then we claim this is the required atlas. Indeed, if $\underline{I} \rightarrow [X/G]$ corresponds to a G_T -bundle $\pi: P \rightarrow T$ and $f: P \rightarrow X$, then one can easily show a cartesian diagram

$$\begin{array}{ccc} P & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow \\ \underline{I} & \longrightarrow & [X/G] \end{array}$$

Since $P \rightarrow T$ is étale & surjective, we're done

Example - ii) Let $X = \text{Spec}(A)$, for a finite type k -algebra A and G be a finite group acting on A . From the fact that G -bundles

on $\text{Spec}(\bar{k})$, for an algebraic closure $k \hookrightarrow \bar{k}$, are trivial, we conclude that

$$[\text{Spec}(A)/G](\bar{k}) = \left\{ G\text{-equivariant maps} \atop G \rightarrow \text{Spec}(A)(\bar{k}) \right\}$$

In particular,

$$\left\{ \begin{array}{l} \text{Isom. classes} \\ \text{in } [\text{Spec}(A)/G](\bar{k}) \end{array} \right\} \xleftrightarrow{|\cdot|} \text{Spec}(A)(\bar{k})/G$$

(orbits of G -
action)

But we know from the work of Mumford that

$$\text{Spec}(A^G)(\bar{k}) = \text{Spec}(A)(\bar{k})/G.$$

$\Rightarrow \text{Spec}(A^G) = \text{Spec}(A)/G$ is a **coarse moduli space** for $[\text{Spec}(A)/G]$ (see [Ch. 6 § 11, Olsson]).

Warning. It is not true in general that $\text{Spec}(A^G)$ represents

$[\mathrm{Spec}(A)/G]$ (happens when the G -action on $\mathrm{Spec}(A)$ has non-trivial isotropy groups). This is showcased in the following simple example: let $G = \mathbb{Z}/(2)$ act on $X = \mathrm{Spec}\left(\frac{k[T, Y]}{(T, Y)^2}\right)$ via $(T, Y) \mapsto (Y, T)$.

Exercise (!) Show that $X/G \cong \mathbb{A}_k^1$.

To show that $[X/G]$ is not representable by $X/G = \mathbb{A}_k^1$, it suffices to show $[X/G]$ is not a smooth stack. Indeed, by what we saw previously, the atlas $\underline{X} \rightarrow [X/G]$ is an étale map. But since $\underline{X}$ is not smooth (being union of two lines), $[X/G]$ is also not smooth.

(ii) The stack of elliptic curves $\mathcal{M}_{1,1}$ is also a DM stack

and admits the coarse moduli space $\mathbb{H}/SL_2(\mathbb{Z})$ (see Talk 1) via

the j -invariant of elliptic curves. (see also [Introduction, Olsson]).

The pattern above is encapsulated by the following deep result, which will not be used by us in this seminar.

Theorem (Keel-Mori). A DM-stack with finite diagonal (for instance, a separated stack) admits a coarse moduli space.

§3. The stack $\mathcal{E}ll_{\mathcal{D}, I}$

Let D be a central division algebra of dimension d^2 over $F = \mathbb{F}_q(t)$. Let $X = \mathbb{P}_{\mathbb{F}_q}^1$ and $\mathcal{D}$ be the locally free $\mathcal{O}_X$ -algebra s.t.

$$D_\eta = D \text{ (generic fiber) and}$$

$$D_x = D \otimes \widehat{\mathcal{O}}_{X,x} \subset F_x \text{ (} x \in X \text{)}$$

is a maximal order.

Recall the definition of a
 D -elliptic sheaf $\underline{\Sigma} = (\Sigma_i, j_i, t_i)_{i \in \mathbb{Z}}$
 over an $\mathbb{F}_q$ -scheme S .

(for the full definition see the
 notes of Talk 2) $\div$ Σ_i are locally
 free $\mathcal{O}_{X \times S}$ -modules of rank
 d^2 with an action by D s.t.

$$(i) \quad \begin{array}{ccc} \Sigma_i & \xrightarrow{j_i} & \Sigma_{i+1} \\ \uparrow t_{i-1} & & \uparrow t_i \\ {}^{\tau}\Sigma_{i-1} & \xrightarrow{{}^{\tau}j_{i-1}} & {}^{\tau}\Sigma_i \end{array} \quad \text{commutes}$$

$$(ii) \quad \Sigma_i \xrightarrow{j_i} \Sigma_{i+1} \longrightarrow \dots \longrightarrow \Sigma_{i+d} = \Sigma_i(\infty)$$

is the canonical injection;

(iii) Condition on the pole ∞

(iv) Condition on the zero's

(v) for all geometric pts $\{s\} \rightarrow S$,
 $\chi(\mathcal{E}_0|_{X \times \{s\}}) \in [0, d^2)$ (normalization condition)

(vi) $\underline{\mathcal{E}}$ is special.

For appropriate ideals $I \subsetneq \mathbb{F}_q[t]$,
we can put level I -structure
on $\underline{\mathcal{E}}$. Consequently this gives a

$$\text{CFG } \mathcal{E}ll_{\mathcal{D}, I} \rightarrow (\text{Sch } \mathbb{F}_q)_{\text{fppf}}.$$

Using descent for vector bundles,
it is not hard to show that
this is an fppf stack. By
abuse of notation, we denote this
stack just by $\mathcal{E}ll_{\mathcal{D}, I}$.

Let $X' = X \setminus \{\infty\} \cup \mathcal{R}$.

THEOREM ([LRS]) $\mathcal{E}ll_{\mathcal{D}, I}$ is a DM-
stack which is smooth of relative
dimension $d-1$ over $X' \setminus I$. Moreover,

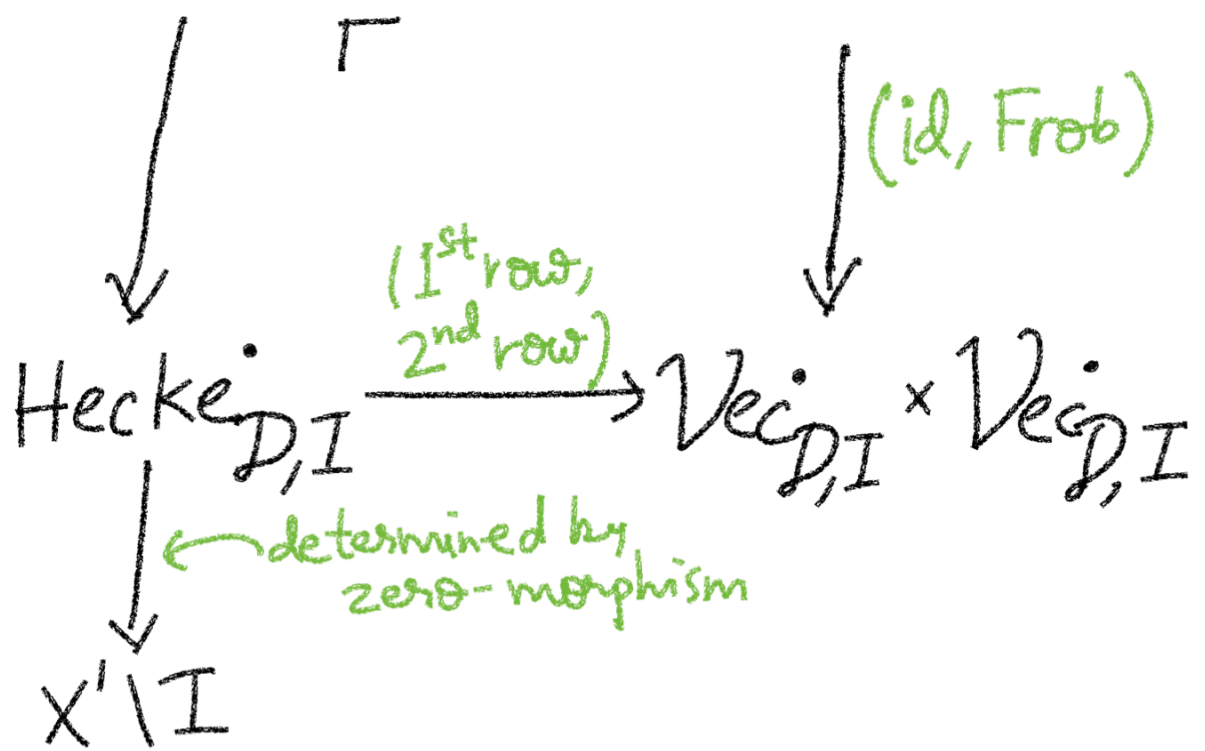
if $I \neq \emptyset$, then $\mathcal{E}ll_{\mathcal{D}, I}$ is representable by a smooth, projective scheme over $X' \setminus I$.

Remark. Note that [LRS] do not impose the conditions (v), (vi) above for a $\mathcal{D}$ -elliptic sheaf. From the work of Spiess, (vi) is equivalent to (iv), and (v) being a normalization condition, we get $\mathcal{E}ll_{\mathcal{D}, I}$ is the quotient by $\mathbb{Z}$ of the corresponding stack in [LRS]. The proof of the above theorem seems to work even after the quotient by $\mathbb{Z}$.

Sketch of a proof

The key to the proof is the following Cartesian diagram of Stacks:

$$\mathcal{E}ll_{\mathcal{D}, I} \longrightarrow \mathcal{V}ec_{\mathcal{D}, I}$$



where

(i) $Vec_{D,I}$ classifies $\{\Sigma_i \xrightarrow{j_i} \Sigma_{i+1}\}_{i \in \mathbb{Z}}$, as in the def'n of D -ell.sheaves, together with conditions (ii) & (iii) and level I -structure.

(ii) $Hecke_{D,I}$ classifies $\left\{ \begin{array}{ccc} \Sigma_i & \xrightarrow{j_i} & \Sigma_{i+1} \\ \uparrow t_H & & \uparrow t_1 \\ \Sigma'_{i-1} & \xrightarrow{j'_{i-1}} & \Sigma'_i \end{array} \right\}$

together with (ii), (iii), (iv) and level I -structure.

Step 1. Let Vec_I^{st} denote the stack classifying rank d^2 vector bundles E which are Seshadri-I-

stable, i.e. $\mathcal{E}$ has a level
 I -structure and for all
subbundles $\mathcal{F} \subset \mathcal{E}$,

$$\frac{\deg(\mathcal{F}) - \deg(I)}{\operatorname{rk}(\mathcal{F})} < \frac{\deg(\mathcal{E}) - \deg(I)}{\operatorname{rk}(\mathcal{E})}$$

([Seshadri, 4.I Definition 2]) If $\deg(I) > 0$
then $\operatorname{Vec}_I^{\operatorname{st}}$ is representable by
a disjoint union of smooth
quasi-projective schemes.

Let now $\operatorname{Ell}_{\mathcal{D}, I}^{\operatorname{st}}$ denote $\mathcal{D}$ -ell.
sheaves $(\underline{\mathcal{E}})$ s.t. $\mathcal{E}_0$ is Seshadri- I -
stable. Then we can conclude
using the following, as well as the
Cartesian diagram above that
 $\operatorname{Ell}_{\mathcal{D}, I}^{\operatorname{st}}$ is a disjoint union of
smooth quasi-projective schemes
if $\deg(I) > 0$.

- (a) $\text{Vec}_{D,I} \rightarrow \text{Vec}_I$ is representable and affine;
- (b) $\text{Vec}_{D,I}$ is smooth;
- (c) $\text{Vec}_{D,I}^\bullet \rightarrow \text{Vec}_{D,I}$ is representable and smooth;
- (d) $\text{Hecke}_{D,I} \xrightarrow[\text{first row}]{(0\text{-mor})} (X' \setminus I) \times \text{Vec}_{D,I}^\bullet$ is representable and smooth of dim'n $d-1$.

Lemma $\div$ Let S, U, V be $\mathbb{F}_q$ -schemes and $\alpha: V \rightarrow U \times U, f: V \rightarrow S$ be $\mathbb{F}_q$ -morphisms. Form the cartesian square $\div$

If $(f, \text{pr}_1, \alpha): V \rightarrow S \times U$ is smooth of dim'n n , then g is smooth of dim'n n .

$$\begin{array}{ccc}
 W & \xrightarrow{\beta} & U \\
 j \downarrow & \lrcorner & \downarrow (\text{id}, \text{Frob}) \\
 V & \xrightarrow{\alpha} & U \times U \\
 \downarrow & & \\
 S & &
 \end{array}$$

Step 2 $\div$ The idea is to cover

$\mathcal{E}ll_{D,I}$ by finite quotients
of $\mathcal{E}ll_{D,J}^{st}$ for $J \not\supseteq I$. Consider
an ideal $J \not\supseteq I$ s.t. $\deg(J) > 0$.

Let $r_{J,I} : \mathcal{E}ll_{D,J} \rightarrow \mathcal{E}ll_{D,I}$

be the quotient of the level
 J -str. by I . This is a

$G_{I,J} := \text{Ker}((\mathcal{O}_{D/J})^* \rightarrow (\mathcal{O}_{D/I})^*)$ -

torsor. Hence, $\mathcal{E}ll_{D,J}^{st} \subset \mathcal{E}ll_{D,I}$

being stable under the action of
 $G_{I,J}$, we get a smooth, open,

DM-substack $[\mathcal{E}ll_{D,J}^{st}/G_{I,J}] \subset$

$\mathcal{E}ll_{D,I}$. Since any vector bundle
is Seshadri- J -stable for $\deg(J) > 0$,
such finite quotient stacks cover
 $\mathcal{E}ll_{D,I}$ and we're done. $\square$

§4. REFERENCES

[OLSSON] Algebraic spaces & Stacks.

[LRS] D-elliptic sheaves and the local Langlands correspondence.

[Seshadri] Fibrés vectoriels sur les courbes algébriques.

